Module 2: Two-Dimensional Electrostatics with Defect-Dependent Space Charge

Derivation-Focused Physics Note for `RadiationEffects_proj2`

Electrostatic bridge between evolving defect populations and carrier transport in silicon

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Purpose of this note. This note develops the electrostatic module that connects radiation-induced defect populations to the internal electric field structure of a silicon device. The goal is to start from Gauss's law and the electrostatic limit of Maxwell's equations, derive Poisson's equation, then show carefully how electrons, holes, dopants, and charged defect populations combine into the space-charge density that drives the potential and electric field solved in Module 2. This module is the electrostatic bridge between the defect-evolution physics of Module 3 and the carrier-transport physics of Module 5. The note also derives the finite-element weak form used by the MATLAB implementation, including the triangular mesh, shape functions, element matrix, source vector, boundary-condition handling, and electric-field postprocessing.

Contents

Symbols and acronyms used in this document

Symbol	Meaning	Typical units
ϕ	Electrostatic potential	V

Symbol	**Meaning**	**Typical units**
$\mathbf{E}$	Electric field	V/m
$\mathbf{D}$	Electric displacement field	$\mathrm{C/m^2}$
ρ	Total space-charge density	$\mathrm{C/m^3}$
ρ_f	Free-charge density in Gauss's law	$\mathrm{C/m^3}$
ρ_{def}	Charge density due to defects	$\mathrm{C/m^3}$
ε, $\varepsilon_{\mathrm{Si}}$	Permittivity, silicon permittivity	F/m
q	Elementary charge magnitude	C
n	Electron concentration	$\mathrm{m^{-3}}$
p	Hole concentration	$\mathrm{m^{-3}}$
N_D^+	Ionized donor concentration	$\mathrm{m^{-3}}$
N_A^-	Ionized acceptor concentration	$\mathrm{m^{-3}}$
C_i	Concentration of defect species i	$\mathrm{m^{-3}}$
z_i	Effective charge number of defect species i	dimensionless
V	Vacancy defect concentration, when used as an explicit defect species	$\mathrm{m^{-3}}$
I	Self-interstitial defect concentration, when used as an explicit defect species	$\mathrm{m^{-3}}$
V_2	Divacancy defect concentration, when used as an explicit defect species	$\mathrm{m^{-3}}$
VO	Vacancy-oxygen complex concentration, when used as an explicit defect species	$\mathrm{m^{-3}}$
x, y	Cartesian spatial coordinates	m
L	Characteristic device length scale	m
ϕ_0	Characteristic potential scale	V
E_0	Characteristic electric-field scale	V/m
N_i, N_j	Finite-element shape functions associated with nodes i and j	dimensionless
Φ_i, Φ_j	Nodal electrostatic potential values in the FEM discretization	V
$\mathbf{\Phi}$	Vector of nodal electrostatic potentials	V

Symbol	Meaning	Typical units
K, K_{ij}	Global electrostatic stiffness matrix and its matrix entry	F in 3D; F/m in 2D per-unit-depth form
$\boldsymbol{b}$, b_i	Global FEM source vector and its nodal entry	C in 3D; C/m in 2D per-unit-depth form
U, U_h	Continuous and finite-element electrostatic field energy	J

1 Purpose of Module 2 and its role in the project

The overall project aims to connect a radiation event in silicon to eventual device degradation. The Geant4 module provides the deposited-energy map, Module 3 evolves the resulting defect population, Module 4 supplies the thermal field, and Module 5 solves carrier transport. Module 2 sits between the material-damage description and the electrical-transport description. Its job is to determine how charged defects and ionized dopants reshape the electrostatic potential and electric field inside the silicon device.

This matters because carrier motion is not governed only by concentration gradients. It is also governed by the internal electric field. If radiation-induced defects add trap charge, compensate dopants, or otherwise modify the net space-charge density, then the potential profile $\phi(x, y)$ changes, the electric field $\mathbf{E}(x, y)$ changes, and therefore the carrier transport problem in Module 5 changes.

Thus Module 2 is the electrostatic translation layer:

$$\text{defect population} \longrightarrow \text{space charge} \longrightarrow \text{electrostatic potential} \longrightarrow \text{electric field.} \tag{1.1}$$

2 Starting point: electrostatics from Maxwell's equations

2.1 Gauss's law in matter

The correct starting point is Gauss's law,

$$\nabla \cdot \mathbf{D} = \rho_f, \tag{2.1}$$

where $\mathbf{D}$ is the electric displacement field and ρ_f is the free-charge density. In a linear dielectric,

$$\mathbf{D} = \varepsilon \mathbf{E}. \tag{2.2}$$

For silicon treated as an isotropic medium with effective permittivity ε, this becomes

$$\nabla \cdot (\varepsilon \mathbf{E}) = \rho. \tag{2.3}$$

Here we use ρ for the total charge density that enters the device-scale Poisson problem. At the module level, that charge density includes mobile carriers, ionized dopants, and charged defects.

2.2 Electrostatic limit

For the present module we assume the electrostatic limit. That means magnetic induction effects are neglected and the electric field is derivable from a scalar potential:

$$\nabla \times \mathbf{E} = 0, \qquad \mathbf{E} = -\nabla \phi. \tag{2.4}$$

Substituting into Gauss's law gives

$$\nabla \cdot (\varepsilon(-\nabla \phi)) = \rho, \tag{2.5}$$

or equivalently,

$$\nabla \cdot (\varepsilon \nabla \phi) = -\rho. \tag{2.6}$$

This is the general Poisson equation for electrostatics in an inhomogeneous dielectric.

2.3 Why Poisson's equation is the right equation here

Poisson's equation expresses a direct local statement: curvature in the electrostatic potential is produced by charge density. If the charge density is positive in a region, the potential bends one way; if negative, it bends the opposite way. In a device, the field is not imposed from nowhere. It is generated self-consistently by the charge distribution plus the boundary conditions set by contacts and external bias.

This is why Module 2 must exist separately from Module 5. Carrier transport needs the electric field, but the electric field itself depends on the charge state of the device.

3 Building the semiconductor space-charge density

3.1 Charge contributions in a semiconductor

In a semiconductor, the charge density is not just a single species. It is a sum of several contributions:

$$\rho = \rho_{\text{holes}} + \rho_{\text{electrons}} + \rho_{\text{dopants}} + \rho_{\text{defects}}. \tag{3.1}$$

Each term must be assigned the correct sign.

- Holes carry charge $+q$, so their contribution is $+qp$.
- Electrons carry charge $-q$, so their contribution is $-qn$.
- Ionized donors are positively charged when ionized, giving $+qN_D^+$.
- Ionized acceptors are negatively charged when ionized, giving $-qN_A^-$.
- Charged defects contribute an additional term ρ_{def}.

Therefore,

$$\rho = q(p - n + N_D^+ - N_A^-) + \rho_{\text{def}}. \tag{3.2}$$

This is the central charge-density relation for Module 2.

3.2 Physical interpretation of each contribution

The signs are worth pausing on because they are easy to write mechanically and easy to misunderstand physically.

A hole is the absence of an electron in a valence-band state. At the continuum device level it behaves as a mobile positive carrier. An electron is a mobile negative carrier. Ionized donors are donor atoms that have donated an electron to the conduction process and are left positively charged. Ionized acceptors are acceptor atoms that have captured an electron and therefore are negatively charged.

Radiation-induced defects complicate this familiar picture because they can act like trap centers with charge states that depend on local occupancy and local electrochemical conditions. Thus ρ_{def} is not merely a bookkeeping term. It is the place where material damage enters the electrostatics.

4 How defect populations enter the electrostatic problem

4.1 Defect species and charge states

Suppose Module 3 evolves several defect concentrations,

$$C_1(x, y, t),\ C_2(x, y, t),\ \ldots,\ C_M(x, y, t). \tag{4.1}$$

In this notation, a *defect species* means a radiation-induced material-defect population, not a mobile carrier or an intentional dopant. Thus C_i does *not* denote the electron concentration n, the hole concentration p, the ionized donor concentration N_D^+, or the ionized acceptor concentration N_A^-. Those carrier and dopant contributions already appear explicitly in the semiconductor space-charge density. The role of the C_i fields is narrower: they represent damage-related populations supplied by Module 3 and converted by Module 2 into an additional defect-charge density.

Examples of possible defect species in irradiated silicon include isolated vacancies, self-interstitials, and defect complexes. A representative mapping is

Example field	**Physical meaning**	**Charge factor**
C_V	Concentration of vacancy defects V	z_V
C_I	Concentration of self-interstitial defects I	z_I
C_{V_2}	Concentration of divacancy complexes V_2	z_{V_2}
C_{VO}	Concentration of vacancy-oxygen complexes VO	z_{VO}
$C_{\mathrm{trap},k}$	Concentration of an effective charged trap family k	$z_{\mathrm{trap},k}$

This table is illustrative rather than exhaustive. The actual list of species depends on how detailed Module 3 is. A simple reduced model may keep only one effective defect concentration C, while a richer defect-kinetics model may separately evolve vacancies, interstitials, divacancies, impurity-defect complexes, and trap families.

Each defect species may possess one or more possible charge states. At the reduced project level, the cleanest way to map defects into electrostatics is to assign each defect species an effective average charge number z_i and write

$$\rho_{\mathrm{def}} = q\sum_{i=1}^{M} z_i C_i. \tag{4.2}$$

If $z_i > 0$, the species contributes positive charge on average. If $z_i < 0$, it contributes negative charge on average. If $z_i = 0$, the species is electrostatically neutral in the reduced model. The parameter z_i is dimensionless; the charge carried per defect is qz_i, not z_i alone.

4.1.1 Why a vacancy is not automatically a positive charge

It is important not to confuse a vacancy defect with a semiconductor hole. A hole is a missing valence-band electron and is represented by the mobile carrier concentration p. A vacancy, by contrast, is a missing silicon atom on a lattice site. When radiation displaces a silicon atom from its lattice position, the primary structural defect is a vacancy-interstitial pair, not automatically a mobile hole.

A silicon atom in a crystal is not simply a positive nucleus sitting at a point. It is part of a covalent bonding network. Removing or displacing that atom changes the local bonding geometry. Neighboring atoms may be left with dangling bonds, the lattice may relax around the missing atom, and localized defect-induced electronic states may appear in the silicon band gap. Those localized states can capture or emit electrons depending on the local Fermi level, temperature, and carrier populations. Therefore the same structural defect may appear in different charge states, for example

$$V^{+}, \qquad V^{0}, \qquad V^{-}, \qquad V^{2-}, \tag{4.3}$$

where V denotes a vacancy-like defect. In this notation, the superscript indicates the net electronic charge state of the defect after accounting for local bond breaking, dangling bonds, lattice relaxation, and occupancy of defect-induced electronic states.

Thus the effective charge factor z_i does not mean that a missing atom is automatically equal to $+q$. Instead, z_i is a reduced charge-state parameter that represents the net electrostatic charge associated with defect species i after microscopic electronic structure and occupancy effects have been collapsed into a device-scale model. For a vacancy-like defect, $z_V = +1$ would represent an average singly positive vacancy population, $z_V = 0$ a neutral vacancy population, $z_V = -1$ a singly negative vacancy population, and $z_V = -2$ a doubly negative vacancy population. In a reduced model, z_i may also be a non-integer effective average if several microscopic charge states are grouped into one continuum species.

This is not yet the most microscopic treatment because a full trap-occupancy model would make z_i depend on Fermi level, temperature, carrier capture rates, carrier emission rates, and defect energy levels in the band gap. But for the present project architecture, the above equation is the correct reduced bridge from Module 3 to Module 2: Module 3 supplies defect concentrations $C_i(x, y, t)$, while Module 2 uses their effective charge factors z_i to compute the electrostatic source term.

4.2 Single-species reduced form

For a first implementation with a single effective defect concentration $C(x, y, t)$ and charge factor z_{def}, we may write

$$\rho_{\text{def}} = q z_{\text{def}} C. \tag{4.4}$$

Then the full charge density becomes

$$\rho = q(p - n + N_D^+ - N_A^- + z_{\text{def}} C). \tag{4.5}$$

This compact equation is often sufficient to explore how defect accumulation or annealing alters depletion and field profiles.

5 Canonical spatial-field input contract

The physical equations above determine what enters Poisson's equation. The software interface should preserve that same physics without forcing the charge distribution into a special low-dimensional shape. The canonical Module 2 input is therefore the complete spatial total-charge field

$$\rho(x, y), \tag{5.1}$$

or, on the FEM mesh,

$$\boxed{\boldsymbol{\rho} = [\rho_1, \rho_2, \ldots, \rho_N]^T} \tag{5.2}$$

with one value in $\mathrm{C/m^3}$ at every active mesh node.

The individual physical contributions may themselves be spatial fields,

$$\boldsymbol{p}, \quad \boldsymbol{n}, \quad \boldsymbol{N}_D^+, \quad \boldsymbol{N}_A^-, \quad \boldsymbol{C}_i, \tag{5.3}$$

and are combined pointwise before the electrostatic solve:

$$\boxed{\boldsymbol{\rho} = q\left(\boldsymbol{p} - \boldsymbol{n} + \boldsymbol{N}_D^+ - \boldsymbol{N}_A^- + \sum_i \boldsymbol{z}_i \odot \boldsymbol{C}_i\right).} \tag{5.4}$$

Here $\odot$ denotes pointwise multiplication. Any scalar concentration is simply interpreted as a uniform field over the mesh.

This means Module 2 does *not* require a list of individual defect coordinates, nor does it require that the charge field be Gaussian. Microscopic defect information must first be coarse-grained into continuum concentrations or total charge density on the Module 2 mesh. Once $\boldsymbol{\rho}$ has been formed, the electrostatic solver only needs that field and the material/boundary data.

Role of the Gaussian model

The previously used single-Gaussian defect description is retained only as a controlled synthetic field generator:

$$(C_{\text{peak}}, x_c, y_c, \sigma_x, \sigma_y) \longrightarrow \boldsymbol{\rho}_{\text{synthetic}}. \tag{5.5}$$

Those five parameters are useful for producing reproducible verification and training cases, but they are no longer the Module 2 input contract. Imported non-Gaussian fields, multi-cluster fields, and future Module 3/Module 5 fields all enter through the same $\boldsymbol{\rho}$ interface.

6 Derivation of the project Poisson equation

Starting from the general electrostatic equation,

$$\nabla \cdot (\varepsilon \nabla \phi) = -\rho, \tag{6.1}$$

and substituting the semiconductor charge density gives

$$\nabla \cdot (\varepsilon \nabla \phi) = -\left[q(p - n + N_D^+ - N_A^-) + \rho_{\text{def}}\right]. \tag{6.2}$$

For silicon with an approximately uniform effective permittivity $\varepsilon = \varepsilon_{\mathrm{Si}}$, we may bring $\varepsilon_{\mathrm{Si}}$ outside the divergence,

$$\varepsilon_{\mathrm{Si}} \nabla^2 \phi = -\rho, \tag{6.3}$$

so that

$$\nabla^2 \phi = -\frac{1}{\varepsilon_{\mathrm{Si}}} \left[q(p - n + N_D^+ - N_A^-) + \rho_{\mathrm{def}} \right]. \tag{6.4}$$

With the effective-charge mapping for defects,

$$\nabla^2 \phi = -\frac{q}{\varepsilon_{\mathrm{Si}}} \left[p - n + N_D^+ - N_A^- + \sum_{i=1}^{M} z_i C_i \right]. \tag{6.5}$$

This is the device-scale electrostatic core of Module 2.

7 Reduction to the two-dimensional project geometry

7.1 Why a two-dimensional model

The project has moved from the original 1D baseline to a 2D architecture for Modules 2–6. The reason is physical rather than cosmetic. A 1D model can represent variation through a device thickness or along a single line, but it cannot represent lateral nonuniformity in energy deposition, defect generation, heating, or bias-driven field shaping. A 2D model is the smallest geometry that can represent both depth variation and lateral structure.

7.2 Cartesian reduction

In Cartesian coordinates (x, y), the Laplacian becomes

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}. \tag{7.1}$$

Therefore the 2D Module 2 equation is

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\frac{1}{\varepsilon_{\mathrm{Si}}} \left[q(p - n + N_D^+ - N_A^-) + \rho_{\mathrm{def}} \right]. \tag{7.2}$$

Or, using the effective defect-charge representation,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\frac{q}{\varepsilon_{\mathrm{Si}}} \left[p - n + N_D^+ - N_A^- + \sum_{i} z_i C_i \right]. \tag{7.3}$$

Once ϕ is solved, the electric field follows from

$$\mathbf{E} = -\nabla\phi = -\hat{\mathbf{x}}\,\frac{\partial\phi}{\partial x} - \hat{\mathbf{y}}\,\frac{\partial\phi}{\partial y}. \tag{7.4}$$

8 Useful limiting cases and physical intuition

8.1 Charge-neutral region

If the net charge density is nearly zero,

$$\rho \approx 0, \tag{8.1}$$

then Poisson's equation reduces to Laplace's equation,

$$\nabla^2\phi = 0. \tag{8.2}$$

In that regime the field is controlled mainly by boundary conditions rather than by internal space charge.

8.2 Uniform fixed space charge

If ρ is approximately uniform, then the potential has quadratic spatial curvature. This is the familiar origin of parabolic potential profiles in depletion approximations. The stronger the magnitude of ρ, the stronger the curvature.

8.3 Effect of positive and negative defect charge

A positively charged defect population acts, from the electrostatic point of view, like an added donor-like contribution. It tends to increase positive space charge and strengthen field curvature in the corresponding direction. A negatively charged defect population acts more like an added acceptor-like contribution. It tends to compensate positive space charge or add negative space charge.

This is why defect charge can change depletion width, barrier shape, leakage pathways, and charge-collection efficiency, even before one solves the carrier transport in detail.

9 Boundary conditions and their physical meaning

The Poisson problem requires boundary conditions. In this project, the most useful classes are the following.

9.1 Dirichlet boundary condition

A Dirichlet condition specifies the potential directly:

$$\phi = \phi_b \quad \text{on a boundary.} \tag{9.1}$$

Physically, this represents a contact or boundary held at a prescribed bias. For example, one contact may be set to SI0V and another to an applied reverse-bias voltage.

9.2 Neumann boundary condition

A Neumann condition specifies the normal electric displacement or, for constant permittivity, the normal derivative of the potential:

$$\frac{\partial \phi}{\partial n} = g \quad \text{on a boundary.} \tag{9.2}$$

If $g = 0$, the normal electric field component vanishes at the boundary. This can represent a symmetry plane or an insulating idealization.

9.3 Mixed physical picture

Many device problems are naturally mixed: contacts use Dirichlet conditions, while symmetry edges or outer truncated computational boundaries may use Neumann conditions. The boundary choices are not mere numerics; they encode the experimental or operational configuration being modeled.

10 Scaling arguments and characteristic magnitudes

Let L be a characteristic device length scale and let ρ_0 be a characteristic charge-density scale. Dimensional reasoning applied to

$$\nabla^2 \phi \sim -\frac{\rho_0}{\varepsilon_{\mathrm{Si}}} \tag{10.1}$$

suggests

$$\frac{\phi_0}{L^2} \sim \frac{\rho_0}{\varepsilon_{\mathrm{Si}}}, \tag{10.2}$$

so the characteristic potential scale is

$$\phi_0 \sim \frac{\rho_0 L^2}{\varepsilon_{\mathrm{Si}}}. \tag{10.3}$$

The associated field scale is then

$$E_0 \sim \frac{\phi_0}{L} \sim \frac{\rho_0 L}{\varepsilon_{\mathrm{Si}}}. \tag{10.4}$$

These relations are useful because they show immediately that the electrostatic impact of defects scales with both the amount of net defect charge and the spatial extent over which that charge persists.

11 How Module 2 couples to the rest of the project

11.1 Input from Module 3

Module 3 provides evolving spatial defect-concentration fields $C_i(x, y, t)$. Module 2 maps those fields into a nodal defect-charge contribution and combines it with any carrier and dopant fields to form the canonical total-charge input $\boldsymbol{\rho}$. Thus diffusion, clustering, or annealing in Module 3 modifies the full spatial source field rather than merely changing a few Gaussian parameters.

11.2 Input from Module 4

Temperature from Module 4 can influence defect charge states and carrier statistics. In a reduced first pass, one may keep the spatial charge map fixed or temperature-independent. In a richer model, temperature can modify the component fields or the effective charge factors before they are combined into $\boldsymbol{\rho}$.

11.3 Output to Module 5

Module 5 needs the electric field. Once Module 2 solves for $\phi(x, y)$, the electric field is obtained from

$$\mathbf{E} = -\nabla\phi. \tag{11.1}$$

That field enters the drift terms in the electron and hole continuity equations. Thus Module 2 provides one of the essential inputs to carrier transport.

12 Finite-element formulation used by the MATLAB implementation

The preceding sections define the continuum electrostatic problem. A MATLAB solver cannot solve the continuum equation directly. It must replace the continuous potential field $\phi(x, y)$ by a finite number of unknown nodal values. This section gives the finite-element method (FEM) form implemented in Module 2.

12.1 Strong form, domain, and boundary split

Let Ω be the two-dimensional silicon cross section and let the boundary be split into a Dirichlet part Γ_D and a Neumann part Γ_N:

$$\partial\Omega = \Gamma_D \cup \Gamma_N, \qquad \Gamma_D \cap \Gamma_N = \emptyset. \tag{12.1}$$

The strong form solved in this module is

$$\nabla \cdot (\varepsilon_{\mathrm{Si}} \nabla \phi) = -\rho \qquad \text{in } \Omega. \tag{12.2}$$

The boundary conditions are

$$\phi = \bar{\phi} \qquad\qquad \text{on } \Gamma_D, \tag{12.3}$$

$$\frac{\partial \phi}{\partial n} = g \qquad\qquad \text{on } \Gamma_N, \tag{12.4}$$

where $\bar{\phi}$ is the prescribed contact potential, n is the outward normal direction, and g is the prescribed normal derivative of potential. The current first implementation uses left and right Dirichlet contacts and homogeneous natural Neumann conditions on the top and bottom boundaries.

12.2 Why the weak form is needed

Directly differentiating an approximate potential twice is unnecessarily restrictive. FEM instead multiplies the governing equation by a test function w and integrates over the domain:

$$\int_\Omega w \, \nabla \cdot (\varepsilon_{\mathrm{Si}} \nabla \phi) \, d\Omega = -\int_\Omega w\rho \, d\Omega. \tag{12.5}$$

Using integration by parts, or the divergence theorem, gives

$$\int_\Gamma w\varepsilon_{\mathrm{Si}} \frac{\partial \phi}{\partial n} \, d\Gamma - \int_\Omega \varepsilon_{\mathrm{Si}} \nabla w \cdot \nabla \phi \, d\Omega = -\int_\Omega w\rho \, d\Omega. \tag{12.6}$$

Rearranging gives the weak form:

$$\boxed{\int_\Omega \varepsilon_{\mathrm{Si}} \nabla w \cdot \nabla \phi \, d\Omega = \int_\Omega w\rho \, d\Omega + \int_{\Gamma_N} w\varepsilon_{\mathrm{Si}} g \, d\Gamma.} \tag{12.7}$$

For the default zero-normal-field boundary, $g = 0$ on Γ_N, the boundary integral vanishes. This is why homogeneous Neumann boundaries are called natural boundary conditions in this formulation: they do not require explicit modification of the global matrix.

12.3 Detailed derivation from weak form to matrix form

This subsection makes explicit the step that is often compressed into one line: how the weak form becomes the algebraic FEM system solved in MATLAB. The starting weak form from the previous subsection is

$$\int_\Omega \varepsilon_{\mathrm{Si}} \nabla w \cdot \nabla \phi \, d\Omega = \int_\Omega w\rho \, d\Omega + \int_{\Gamma_N} w\varepsilon_{\mathrm{Si}} g \, d\Gamma. \tag{12.8}$$

Here w is the test function, ϕ is the electrostatic potential, ρ is the Module 2 space-charge density, and $g = \partial\phi/\partial n$ is the prescribed normal derivative on the Neumann boundary. For the present electrostatic module,

$$\rho = q(p - n + N_D^+ - N_A^-) + \rho_{\mathrm{def}}, \qquad \rho_{\mathrm{def}} = q\sum_i z_i C_i. \tag{12.9}$$

Thus the defect populations enter the FEM solve through the right-hand-side source vector.

12.3.1 Global finite-element approximation

The continuous potential field is replaced by a finite-dimensional approximation:

$$\phi_h(\boldsymbol{x}) = \sum_{j=1}^{N_{\mathrm{node}}} \Phi_j N_j(\boldsymbol{x}), \tag{12.10}$$

where $\phi_h(\boldsymbol{x})$ is the approximate potential field, Φ_j is the unknown nodal potential at global node j, $N_j(\boldsymbol{x})$ is the global shape function associated with node j, and N_{node} is the total number of mesh nodes.

Equation (12.10) is a global expression. It does not mean that every node contributes to the potential at every point. Standard FEM shape functions have local support. Therefore, if $\boldsymbol{x}$ lies inside element e, only the shape functions associated with the nodes of that element are nonzero at that point. The local element form is

$$\phi_h^{(e)}(\boldsymbol{x}) = \sum_{a=1}^{n_{\mathrm{en}}} \Phi_{I_e(a)} N_a^{(e)}(\boldsymbol{x}), \qquad \boldsymbol{x} \in \Omega_e, \tag{12.11}$$

where n_{en} is the number of element nodes, a is a local node index, $I_e(a)$ maps local node a in element e to the corresponding global node index, and $N_a^{(e)}$ is the local shape function. For the linear triangular elements used in Module 2,

$$n_{\mathrm{en}} = 3. \tag{12.12}$$

The spatial gradient of the approximate potential is

$$\nabla\phi_h(\boldsymbol{x}) = \nabla\left(\sum_{j=1}^{N_{\text{node}}} \Phi_j N_j(\boldsymbol{x})\right). \tag{12.13}$$

The nodal values Φ_j are coefficients, not spatial functions, so the gradient acts only on the shape functions:

$$\boxed{\nabla\phi_h(\boldsymbol{x}) = \sum_{j=1}^{N_{\text{node}}} \Phi_j \nabla N_j(\boldsymbol{x}).} \tag{12.14}$$

Inside element e, this becomes

$$\nabla\phi_h^{(e)}(\boldsymbol{x}) = \sum_{a=1}^{n_{\text{en}}} \Phi_{I_e(a)} \nabla N_a^{(e)}(\boldsymbol{x}). \tag{12.15}$$

12.3.2 Galerkin choice of the test function

Module 2 uses a Galerkin FEM discretization. This means that the test functions are chosen from the same space as the potential approximation. For global node i, choose

$$w = N_i. \tag{12.16}$$

Substituting $w = N_i$ and $\phi \approx \phi_h$ into Eq. (12.8) gives

$$\int_\Omega \varepsilon_{\text{Si}} \nabla N_i \cdot \nabla\phi_h \, d\Omega = \int_\Omega N_i \rho \, d\Omega + \int_{\Gamma_N} N_i \varepsilon_{\text{Si}} g \, d\Gamma. \tag{12.17}$$

Now insert Eq. (12.14) into the left-hand side:

$$\int_\Omega \varepsilon_{\text{Si}} \nabla N_i \cdot \left(\sum_{j=1}^{N_{\text{node}}} \Phi_j \nabla N_j\right) d\Omega = \int_\Omega N_i \rho \, d\Omega + \int_{\Gamma_N} N_i \varepsilon_{\text{Si}} g \, d\Gamma. \tag{12.18}$$

Because Φ_j is a constant coefficient with respect to position, the sum may be pulled outside the integral:

$$\sum_{j=1}^{N_{\text{node}}} \Phi_j \int_\Omega \varepsilon_{\text{Si}} \nabla N_i \cdot \nabla N_j \, d\Omega = \int_\Omega N_i \rho \, d\Omega + \int_{\Gamma_N} N_i \varepsilon_{\text{Si}} g \, d\Gamma. \tag{12.19}$$

This equation is now algebraic in the unknown nodal potentials Φ_j.

12.3.3 Definition of the global matrix and source vector

Define the global electrostatic stiffness matrix by

$$\boxed{K_{ij} = \int_\Omega \varepsilon_{\text{Si}} \nabla N_i \cdot \nabla N_j \, d\Omega.} \tag{12.20}$$

Define the global source vector by

$$\boxed{b_i = \int_\Omega N_i \rho \, d\Omega + \int_{\Gamma_N} N_i \varepsilon_{\text{Si}} g \, d\Gamma.} \tag{12.21}$$

With these definitions, Eq. (12.19) becomes

$$\boxed{\sum_{j=1}^{N_{\text{node}}} K_{ij} \Phi_j = b_i, \qquad i = 1, 2, \ldots, N_{\text{node}}.} \tag{12.22}$$

Writing all rows together gives the global matrix equation

$$\boxed{K\boldsymbol{\Phi} = \boldsymbol{b}.} \tag{12.23}$$

Some FEM texts call the right-hand side $\boldsymbol{F}$. In this project, the MATLAB implementation and the surrounding notes use $\boldsymbol{b}$ for the source vector, so the Module 2 algebraic system is written as $K\boldsymbol{\Phi} = \boldsymbol{b}$.

For the default homogeneous Neumann boundary on insulated top and bottom surfaces,

$$g = 0 \qquad \text{on } \Gamma_N, \tag{12.24}$$

so the boundary integral in Eq. (12.21) vanishes:

$$\boxed{b_i = \int_\Omega N_i \rho \, d\Omega.} \tag{12.25}$$

Using the Module 2 charge model, this is

$$\boxed{b_i = \int_\Omega N_i \left[q(p - n + N_D^+ - N_A^-) + \rho_{\text{def}} \right] d\Omega.} \tag{12.26}$$

Equivalently,

$$\boxed{b_i = \int_\Omega q N_i \left[p - n + N_D^+ - N_A^- + \sum_k z_k C_k \right] d\Omega.} \tag{12.27}$$

This expression is the direct mathematical link between radiation-induced charged defects and the FEM right-hand side.

12.3.4 Physical interpretation of the electrostatic stiffness matrix

The matrix entry in Eq. (12.20),

$$K_{ij} = \int_\Omega \varepsilon_{\mathrm{Si}} \nabla N_i \cdot \nabla N_j \, d\Omega, \tag{12.28}$$

is the part of the FEM derivation that is easiest to treat as just an abstract mathematical object. It is more useful to attach a physical meaning to it.

A good first interpretation is

$$\boxed{K\mathbf{\Phi} \quad \text{is the FEM-discretized version of} \quad -\nabla\cdot(\varepsilon_{\mathrm{Si}}\nabla\phi)\,.} \tag{12.29}$$

This statement is correct, but it requires one important refinement. In the strong form, the operator contains the divergence of the gradient of the potential. In the weak form, integration by parts moves one spatial derivative from the potential field onto the test function. Therefore the FEM matrix contains products of first derivatives,

$$\nabla N_i \cdot \nabla N_j, \tag{12.30}$$

rather than a direct second derivative such as $\nabla^2 N_j$. The second-derivative operator is still represented, but it is represented in an integral, weak-form sense.

Shape-function gradients and the electric field. The FEM potential is

$$\phi_h(\mathbf{x}) = \sum_{j=1}^{N_{\mathrm{node}}} \Phi_j N_j(\mathbf{x}). \tag{12.31}$$

Taking the gradient gives

$$\nabla\phi_h(\mathbf{x}) = \sum_{j=1}^{N_{\mathrm{node}}} \Phi_j \nabla N_j(\mathbf{x}). \tag{12.32}$$

The electric field is

$$\mathbf{E}_h(\mathbf{x}) = -\nabla\phi_h(\mathbf{x}), \tag{12.33}$$

so

$$\mathbf{E}_h(\mathbf{x}) = -\sum_{j=1}^{N_{\mathrm{node}}} \Phi_j \nabla N_j(\mathbf{x}). \tag{12.34}$$

Thus N_j tells how nodal potential Φ_j contributes to the potential field, while ∇N_j tells how nodal potential Φ_j contributes to the electric field. This is the first key physical interpretation:

$$\boxed{\nabla N_j \quad \text{is the electric-field pattern, up to a minus sign, produced by a unit potential at node } j.} \tag{12.35}$$

To see this, imagine setting one nodal degree of freedom to unity and all others to zero:

$$\Phi_j = 1, \qquad \Phi_{k\neq j} = 0. \tag{12.36}$$

Then the finite-element potential field is simply

$$\phi_h(\mathbf{x}) = N_j(\mathbf{x}), \tag{12.37}$$

and the associated field-like quantity is

$$-\nabla N_j(\mathbf{x}). \tag{12.38}$$

Therefore the gradient of a shape function is not just a mathematical derivative. It is the local electric-field contribution associated with that nodal basis function.

Meaning of the dot product. The dot product

$$\nabla N_i \cdot \nabla N_j \tag{12.39}$$

measures how much the field pattern associated with node i overlaps with the field pattern associated with node j. If the two shape-function gradients point in similar directions over an element, their dot product is positive. If they point in opposite directions, their dot product is negative. If their supports do not overlap, the product is zero because at least one of the shape-function gradients is zero in that region.

Including the silicon permittivity gives the dielectric-weighted overlap:

$$\boxed{K_{ij} = \text{dielectric-weighted overlap of the electric-field patterns associated with nodes } i \text{ and } j.} \tag{12.40}$$

This is why K_{ij} may be interpreted as an electrostatic coupling between node i and node j. The coupling is nonzero only when the two basis functions share element support. This locality is one of the reasons the global FEM matrix is sparse.

Energy interpretation. The electrostatic field energy stored in a dielectric is

$$U = \frac{1}{2}\int_\Omega \varepsilon_{\mathrm{Si}} |\mathbf{E}|^2 \, d\Omega. \tag{12.41}$$

Using $\mathbf{E} = -\nabla\phi$, this can also be written as

$$U = \frac{1}{2}\int_\Omega \varepsilon_{\mathrm{Si}} |\nabla\phi|^2 \, d\Omega. \tag{12.42}$$

For the FEM approximation,

$$\nabla\phi_h = \sum_j \Phi_j \nabla N_j. \tag{12.43}$$

Therefore the discrete electrostatic energy is

$$U_h = \frac{1}{2}\int_\Omega \varepsilon_{\mathrm{Si}} \left(\sum_i \Phi_i \nabla N_i\right) \cdot \left(\sum_j \Phi_j \nabla N_j\right) \, d\Omega \tag{12.44}$$

$$= \frac{1}{2}\sum_i \sum_j \Phi_i \Phi_j \int_\Omega \varepsilon_{\mathrm{Si}} \nabla N_i \cdot \nabla N_j \, d\Omega. \tag{12.45}$$

Using the definition of K_{ij},

$$\boxed{U_h = \frac{1}{2}\mathbf{\Phi}^T K \mathbf{\Phi}.} \tag{12.46}$$

This gives a second important interpretation:

$$\boxed{K \text{ tells how much electrostatic field energy is produced by a given nodal potential distribution.}} \tag{12.47}$$

The matrix is called a stiffness matrix because it penalizes sharp spatial variations in potential. In structural mechanics, stiffness penalizes displacement gradients. Here, electrostatic stiffness penalizes potential gradients. Large potential changes over short distances produce large $|\nabla\phi|$, large electric field magnitude $|\mathbf{E}|$, and therefore large electrostatic field energy.

One-dimensional element example. The same interpretation becomes especially clear in a one-dimensional linear element. Consider an element of length h with two nodes. The linear shape functions are

$$N_1(x) = 1 - \frac{x}{h}, \qquad N_2(x) = \frac{x}{h}. \tag{12.48}$$

Their derivatives are

$$\frac{dN_1}{dx} = -\frac{1}{h}, \qquad \frac{dN_2}{dx} = \frac{1}{h}. \tag{12.49}$$

For a one-dimensional element with cross-sectional area A, the local stiffness entries are

$$K_{ab}^{(e)} = \int_0^h \varepsilon_{\mathrm{Si}} A \frac{dN_a}{dx}\frac{dN_b}{dx} \, dx. \tag{12.50}$$

Thus

$$K_{11}^{(e)} = \int_0^h \varepsilon_{\mathrm{Si}} A \left(-\frac{1}{h}\right)\left(-\frac{1}{h}\right) dx = \frac{\varepsilon_{\mathrm{Si}} A}{h}, \tag{12.51}$$

$$K_{22}^{(e)} = \int_0^h \varepsilon_{\mathrm{Si}} A \left(\frac{1}{h}\right)\left(\frac{1}{h}\right) dx = \frac{\varepsilon_{\mathrm{Si}} A}{h}, \tag{12.52}$$

$$K_{12}^{(e)} = \int_0^h \varepsilon_{\mathrm{Si}} A \left(-\frac{1}{h}\right)\left(\frac{1}{h}\right) dx = -\frac{\varepsilon_{\mathrm{Si}} A}{h}. \tag{12.53}$$

Therefore

$$\boxed{K^{(e)} = \frac{\varepsilon_{\mathrm{Si}} A}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.} \tag{12.54}$$

This has the same algebraic form as a spring stiffness matrix. The corresponding element energy is

$$\boxed{U^{(e)} = \frac{1}{2} \frac{\varepsilon_{\mathrm{Si}} A}{h} (\Phi_2 - \Phi_1)^2 .} \tag{12.55}$$

Only the potential difference matters. If $\Phi_1 = \Phi_2$, then $\nabla \phi_h = 0$, the electric field is zero, and the stored field energy is zero. If $\Phi_2 - \Phi_1$ is large, then

$$E_x = -\frac{\Phi_2 - \Phi_1}{h} \tag{12.56}$$

is large in magnitude, and the stored field energy increases. In this simple case, the stiffness matrix literally penalizes nodal voltage differences.

Why neighboring off-diagonal entries are often negative. For neighboring nodes, the corresponding shape functions often slope in opposite directions inside the same element. In the one-dimensional example,

$$\frac{dN_1}{dx} < 0, \qquad \frac{dN_2}{dx} > 0, \tag{12.57}$$

so

$$\frac{dN_1}{dx} \frac{dN_2}{dx} < 0. \tag{12.58}$$

This gives

$$K_{12}^{(e)} < 0. \tag{12.59}$$

The negative off-diagonal entry does not mean negative physical energy. Rather, it means neighboring nodal potentials are coupled through the electric field between them. The full quadratic form $\boldsymbol{\Phi}^T K \boldsymbol{\Phi}$ remains nonnegative for the electrostatic problem with appropriate boundary conditions. The matrix favors smooth potential fields unless the charge source $\boldsymbol{b}$ and the boundary conditions require curvature.

Relation back to Gauss's law. The strong form used in Module 2 is

$$-\nabla \cdot (\varepsilon_{\mathrm{Si}} \nabla \phi) = \rho. \tag{12.60}$$

The weak form is

$$\int_\Omega \varepsilon_{\mathrm{Si}} \nabla \phi \cdot \nabla w \, d\Omega = \int_\Omega \rho w \, d\Omega + \int_{\Gamma_N} \varepsilon_{\mathrm{Si}} g w \, d\Gamma. \tag{12.61}$$

Choosing $w = N_i$ gives

$$\int_\Omega \varepsilon_{\mathrm{Si}} \nabla \phi_h \cdot \nabla N_i \, d\Omega = \int_\Omega \rho N_i \, d\Omega + \int_{\Gamma_N} \varepsilon_{\mathrm{Si}} g N_i \, d\Gamma. \tag{12.62}$$

The left-hand side is

$$\sum_j K_{ij} \Phi_j. \tag{12.63}$$

Therefore

$$\boxed{(K\boldsymbol{\Phi})_i = \int_\Omega \varepsilon_{\mathrm{Si}} \nabla \phi_h \cdot \nabla N_i \, d\Omega.} \tag{12.64}$$

This quantity is the dielectric-weighted field response tested around node i. The right-hand side

$$b_i = \int_\Omega \rho N_i \, d\Omega + \int_{\Gamma_N} \varepsilon_{\mathrm{Si}} g N_i \, d\Gamma \tag{12.65}$$

is the charge and boundary-flux forcing assigned to node i through the weighting function N_i. Therefore the row equation

$$\sum_j K_{ij} \Phi_j = b_i \tag{12.66}$$

means

$$\boxed{\begin{aligned}&\text{field-flux response associated with node } i \\ &\qquad = \text{space-charge and boundary-flux forcing assigned to node } i.\end{aligned}} \tag{12.67}$$

This is the discrete weak-form statement of Gauss's law.

Module 2 interpretation. For Module 2, the source term is built from mobile carriers, ionized dopants, and charged defects:

$$\rho = q\left(p - n + N_D^+ - N_A^-\right) + \rho_{\mathrm{def}} = q\left(p - n + N_D^+ - N_A^- + \sum_k z_k C_k\right). \tag{12.68}$$

The finite-element equation is

$$K\boldsymbol{\Phi} = \boldsymbol{b}, \tag{12.69}$$

with

$$K_{ij} = \int_\Omega \varepsilon_{\mathrm{Si}} \nabla N_i \cdot \nabla N_j \, d\Omega \tag{12.70}$$

and, for insulating Neumann boundaries,

$$b_i = \int_\Omega q N_i \left(p - n + N_D^+ - N_A^- + \sum_k z_k C_k\right) d\Omega. \tag{12.71}$$

Thus $\boldsymbol{b}$ contains the dopant and radiation-defect charge forcing, while K contains the dielectric geometry and field-coupling response. More compactly,

$$\boxed{K_{ij} = \text{how strongly the potential at node } j \text{ influences the electric-field balance tested at node } i.} \tag{12.72}$$

Equivalently,

$$\boxed{K_{ij} = \text{electrostatic coupling between node } i \text{ and node } j,} \tag{12.73}$$

weighted by the silicon permittivity, the mesh geometry, and the overlap of the corresponding shape-function gradients.

The most useful mental model is therefore

$$N_j \longrightarrow \text{how node } j \text{ contributes to potential,} \tag{12.74}$$

$$\nabla N_j \longrightarrow \text{how node } j \text{ contributes to electric field,} \tag{12.75}$$

$$\nabla N_i \cdot \nabla N_j \longrightarrow \text{overlap of the field patterns from nodes } i \text{ and } j, \tag{12.76}$$

$$K_{ij} \longrightarrow \text{dielectric-weighted field-pattern overlap between nodes } i \text{ and } j. \tag{12.77}$$

Finally,

$$\boxed{K\mathbf{\Phi} \quad \text{is the discrete weak-form Gauss-law operator.}} \tag{12.78}$$

It corresponds to $-\nabla \cdot (\varepsilon_{\text{Si}} \nabla \phi)$, but in an integral finite-element sense rather than as a pointwise second derivative.

12.3.5 Element-level form and assembly

Although Eqs. (12.20) and (12.21) are written as global integrals, the code computes them element by element. For element e, the local matrix is

$$\boxed{K_{ab}^{(e)} = \int_{\Omega_e} \varepsilon_{\text{Si}} \nabla N_a^{(e)} \cdot \nabla N_b^{(e)} \, d\Omega,} \tag{12.79}$$

and the local source vector is

$$\boxed{b_a^{(e)} = \int_{\Omega_e} N_a^{(e)} \rho \, d\Omega + \int_{\Gamma_N \cap \partial\Omega_e} N_a^{(e)} \varepsilon_{\text{Si}} g \, d\Gamma.} \tag{12.80}$$

For the default insulating Neumann boundary, the second term in Eq. (12.80) is zero.

The local element quantities are inserted into the global arrays through the element connectivity map. If local node a of element e corresponds to global node $A = I_e(a)$, and local node b corresponds to global node $B = I_e(b)$, then assembly means

$$K_{AB} \leftarrow K_{AB} + K_{ab}^{(e)}, \qquad b_A \leftarrow b_A + b_a^{(e)}. \tag{12.81}$$

Repeating this over all elements gives the global system $K\boldsymbol{\Phi} = \boldsymbol{b}$.

12.3.6 Dirichlet contact potentials

The matrix equation above is the assembled equation before strongly imposing contact potentials. If the contact nodes are prescribed by

$$\phi = \bar{\phi} \qquad \text{on } \Gamma_D, \tag{12.82}$$

then the corresponding entries of $\boldsymbol{\Phi}$ are known. Splitting the nodal potentials into free unknown nodes and prescribed Dirichlet nodes gives

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi}_f \\ \boldsymbol{\Phi}_d \end{bmatrix}, \tag{12.83}$$

and the assembled system may be written as

$$\begin{bmatrix} K_{ff} & K_{fd} \\ K_{df} & K_{dd} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Phi}_f \\ \boldsymbol{\Phi}_d \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_f \\ \boldsymbol{b}_d \end{bmatrix}. \tag{12.84}$$

The equation for the unknown free nodes is therefore

$$\boxed{K_{ff}\boldsymbol{\Phi}_f = \boldsymbol{b}_f - K_{fd}\boldsymbol{\Phi}_d.} \tag{12.85}$$

This is the reduced system implied by imposing fixed contact potentials. The MATLAB implementation enforces the same idea by directly modifying the matrix rows and right-hand side in `apply_dirichlet_bc.m`.

12.3.7 Physical interpretation of the matrix equation

The compact equation

$$K\boldsymbol{\Phi} = \boldsymbol{b} \tag{12.86}$$

means

$$\text{dielectric/mesh response} \times \text{nodal potential values} = \text{space-charge forcing.} \tag{12.87}$$

The left-hand side contains $\varepsilon_{\mathrm{Si}}$ and gradients of shape functions, so it describes how the silicon dielectric and mesh geometry resist spatial variation in potential. The right-hand side contains the space charge,

$$\rho = q(p - n + N_D^+ - N_A^-) + q\sum_i z_i C_i, \tag{12.88}$$

so it is where dopants, carriers, and charged radiation-induced defects enter the electrostatic solve. After solving for $\mathbf{\Phi}$, the continuous approximate field is reconstructed through $\phi_h(\boldsymbol{x})$, and the electric field follows from

$$\mathbf{E}_h(\boldsymbol{x}) = -\nabla\phi_h(\boldsymbol{x}). \tag{12.89}$$

Therefore the Module 2 chain is

$$\boxed{\{C_i, z_i\} \longrightarrow \rho_{\text{def}} \longrightarrow \rho \longrightarrow \boldsymbol{b} \longrightarrow \mathbf{\Phi} \longrightarrow \phi_h(\boldsymbol{x}) \longrightarrow \mathbf{E}_h(\boldsymbol{x}).} \tag{12.90}$$

12.4 Triangular mesh and nodal degrees of freedom

The implemented solver begins with a rectangular computational domain and splits it into linear triangular elements. The mesh consists of

- nodal coordinates $\boldsymbol{x}_a = (x_a, y_a)$,
- triangular elements defined by three node indices,
- boundary node sets for the left, right, bottom, and top boundaries.

The unknown degree of freedom is one scalar electrostatic potential per node:

$$\mathbf{\Phi} = \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_N \end{bmatrix}^T, \tag{12.91}$$

where N is the total number of mesh nodes.

This is the electrostatic analogue of the displacement vector in a structural FEM problem. In mechanics, the nodal unknown may be displacement. In Module 2, the nodal unknown is the electric potential.

12.5 Linear triangular shape functions

Within each triangular element e, the potential is approximated as a weighted sum of nodal potentials:

$$\phi^{(e)}(x, y) \approx \sum_{a=1}^{3} N_a(x, y)\phi_a^{(e)}. \tag{12.92}$$

The functions $N_a(x, y)$ are the element shape functions. For a linear triangle they are first-order polynomials of the form

$$N_a(x, y) = \alpha_a + \beta_a x + \gamma_a y. \tag{12.93}$$

They satisfy the interpolation property

$$N_a(\boldsymbol{x}_b) = \delta_{ab}, \tag{12.94}$$

where δ_{ab} is the Kronecker delta. This means shape function N_a is one at node a and zero at the other two local nodes. The practical meaning is simple: the shape functions fill in the potential between nodes.

For a triangle with node coordinates (x_1, y_1), (x_2, y_2), and (x_3, y_3), define its area A_e by

$$2A_e = \det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}. \tag{12.95}$$

The gradients of the shape functions are constant inside a linear triangle:

$$\nabla N_a = \frac{1}{2A_e} \begin{bmatrix} b_a \\ c_a \end{bmatrix}, \tag{12.96}$$

with

$$b_1 = y_2 - y_3, \qquad c_1 = x_3 - x_2, \tag{12.97}$$

$$b_2 = y_3 - y_1, \qquad c_2 = x_1 - x_3, \tag{12.98}$$

$$b_3 = y_1 - y_2, \qquad c_3 = x_2 - x_1. \tag{12.99}$$

Because the gradients are constant, the electric field is also constant inside each linear triangular element:

$$\mathbf{E}^{(e)} = -\nabla \phi^{(e)} = -\sum_{a=1}^{3} \phi_a^{(e)} \nabla N_a. \tag{12.100}$$

12.6 Element stiffness matrix

Choose the Galerkin method, meaning the test functions are chosen from the same shape-function space as the potential approximation. Taking $w = N_a$ and substituting the finite-element approximation gives the element matrix entries

$$K_{ab}^{(e)} = \int_{\Omega_e} \varepsilon_{\mathrm{Si}} \nabla N_a \cdot \nabla N_b \, d\Omega. \tag{12.101}$$

For a linear triangle with constant $\varepsilon_{\mathrm{Si}}$, this integral reduces to

$$K_{ab}^{(e)} = \varepsilon_{\mathrm{Si}} A_e \, \nabla N_a \cdot \nabla N_b. \tag{12.102}$$

Equivalently,

$$K_{ab}^{(e)} = \frac{\varepsilon_{\mathrm{Si}}}{4A_e} \left(b_a b_b + c_a c_b \right). \tag{12.103}$$

This matrix is the electrostatic analogue of a structural stiffness matrix. It does not mean mechanical stiffness; it measures how strongly the energy-like quantity $\int \varepsilon|\nabla\phi|^2 d\Omega$ penalizes changes in potential between connected nodes.

12.7 Element source vector from space charge

The charge density $\rho(x, y)$ acts as the source term. In each element, the source vector is

$$b_a^{(e)} = \int_{\Omega_e} N_a \rho \, d\Omega. \tag{12.104}$$

If ρ is represented by nodal values and linearly interpolated inside the element,

$$\rho^{(e)}(x, y) \approx \sum_{a=1}^{3} N_a(x, y)\rho_a^{(e)}, \tag{12.105}$$

then the consistent source vector is

$$\boldsymbol{b}^{(e)} = \frac{A_e}{12} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} \rho_1^{(e)} \\ \rho_2^{(e)} \\ \rho_3^{(e)} \end{bmatrix}. \tag{12.106}$$

This expression is implemented in `compute_element_source_triangle.m`.

12.8 Global assembly

Each element contributes a local 3×3 matrix $K^{(e)}$ and a local three-entry source vector $\boldsymbol{b}^{(e)}$. Assembly means adding these element contributions into the global matrix and global right-hand side according to the element-to-global node connectivity:

$$K_{AB} \leftarrow K_{AB} + K_{ab}^{(e)}, \qquad b_A \leftarrow b_A + b_a^{(e)}, \tag{12.107}$$

where local node a in element e corresponds to global node A, and local node b corresponds to global node B.

After assembly, the global algebraic system is

$$\boxed{K\boldsymbol{\Phi} = \boldsymbol{b}.} \tag{12.108}$$

This is the discrete FEM version of the Module 2 Poisson equation.

12.9 Boundary-condition insertion

Dirichlet boundary conditions prescribe known potentials at selected nodes. If node i is held at $\phi_i = \bar{\phi}_i$, the corresponding unknown is no longer solved freely. The implementation applies this strongly by modifying the global matrix and right-hand side so that the final linear system directly enforces

$$\Phi_i = \bar{\phi}_i. \tag{12.109}$$

In the default device-like cases, the left boundary is set to one contact potential and the right boundary to another. The top and bottom boundaries are left as homogeneous natural Neumann boundaries unless explicitly changed.

12.10 Electric-field postprocessing

After solving for $\mathbf{\Phi}$, the code computes the electric field in each element using

$$\mathbf{E}^{(e)} = -\sum_{a=1}^{3} \phi_a^{(e)} \nabla N_a. \tag{12.110}$$

Because a linear triangle gives a constant potential gradient inside each element, the element-level field is piecewise constant. For plotting, the implementation also computes a nodal electric field by area-weighted averaging of neighboring element fields.

12.11 MATLAB implementation map

The current implementation is organized as follows:

File	**Role**
`main_module2_electrostatics.m`	Top-level driver for Module 2 test cases and plots.
`default_module2_params.m`	Defines physical constants, geometry defaults, synthetic demonstration parameters, and boundary conditions.
`make_module2_charge_field_2d.m`	Packages and validates the canonical arbitrary nodal total-charge field.
`compose_module2_charge _field_2d.m`	Combines scalar or nodal $p, n, N_D^+, N_A^-, C_{\text{def}}, z_{\text{def}}$ inputs into total nodal ρ.
`generate_module2_gaussian _charge_field_2d.m`	Generates a controlled Gaussian-based synthetic charge field for tests and dataset construction.

File	Role
`build_space_charge_module2_2d.m`	Legacy compatibility wrapper around the synthetic Gaussian field generator.
`make_rectangular_tri_mesh_2d.m`	Builds the rectangular triangular mesh and boundary node sets.
`compute_element_stiffness_triangle.m`	Computes $K^{(e)}$ for a three-node triangular element.
`compute_element_source_triangle.m`	Computes $\boldsymbol{b}^{(e)}$ from nodal charge density.
`assemble_poisson_fem_2d.m`	Assembles the global FEM matrix and source vector.
`apply_dirichlet_bc.m`	Imposes known contact potentials strongly.
`solve_poisson_defect_space_charge_2d.m`	Accepts an explicit nodal charge field, then orchestrates assembly, boundary handling, linear solve, and field calculation.
`solve_module2_nodal_charge_batch_2d.m`	Reuses one FEM operator and factorization for many arbitrary $\boldsymbol{\rho} \to \boldsymbol{\Phi}$ field pairs.
`compute_electric_field_from_potential_2d.m`	Computes element and nodal electric fields from the solved potential.

12.12 What an interviewer may ask about the FEM side

The essential answer is: Module 2 solves a scalar elliptic Poisson equation using a Galerkin finite-element discretization with linear triangular elements. The nodal unknown is electrostatic potential, the element matrix comes from $\int \varepsilon \nabla N_a \cdot \nabla N_b d\Omega$, the source vector comes from $\int N_a \rho d\Omega$, Dirichlet contacts are imposed strongly, homogeneous Neumann boundaries enter naturally, and the electric field is recovered as the negative gradient of the solved potential.

13 Verification logic for Module 2

A derivation-heavy note should also state how the module can be checked. The MATLAB implementation includes the tests below in `matlab/tests/`.

13.1 Uniform zero-charge test

If $\rho = 0$ and consistent constant Dirichlet boundary conditions are imposed, the solution should be constant potential and zero field.

13.2 Linear-potential test

If opposite sides are held at different fixed potentials and the interior charge density is zero, the solution in a rectangular domain should reduce to the corresponding harmonic solution. In a simple one-direction variation, this becomes a linear potential profile.

13.3 Uniform space-charge test

For a known uniform charge density in a simple geometry, the numerical solution should reproduce the expected quadratic potential curvature and linear field variation.

13.4 Defect-charge perturbation test

A localized charged-defect region should create a localized perturbation in the potential and electric field. As the region diffuses or anneals in Module 3, the perturbation should weaken or spread accordingly.

14 What this module captures and what it does not

Module 2 captures the electrostatic consequences of charge rearrangement and defect charge. It does not, by itself, capture full nonequilibrium carrier dynamics, displacement current, or fast electromagnetic wave phenomena. Those belong elsewhere. The module is intentionally the electrostatic layer, not the full semiconductor transient-transport layer.

This distinction is important: Poisson's equation is not the whole device model. It is the field equation inside the larger coupled framework.

15 Summary of the final project equation

The final reduced Module 2 equation for the present project is

$$\nabla \cdot (\varepsilon_{\mathrm{Si}} \nabla \phi) = -\left[q(p - n + N_D^+ - N_A^-) + \rho_{\mathrm{def}} \right], \tag{15.1}$$

with

$$\rho_{\text{def}} = q \sum_i z_i C_i, \tag{15.2}$$

and, for the 2D Cartesian geometry with uniform silicon permittivity,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\frac{q}{\varepsilon_{\text{Si}}} \left[p - n + N_D^+ - N_A^- + \sum_i z_i C_i \right]. \tag{15.3}$$

The electric field is then

$$\mathbf{E} = -\nabla \phi. \tag{15.4}$$

This is the equation set that turns evolving radiation-induced material damage into an evolving internal field structure.

16 References

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